

ECA0905-DIGITAL SIGNAL PROCESSING

DATE:24-07-2026

Experiment 1: Design of Low Pass Filter using Windowing Techniques

4..1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

PROGRAM

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('--- Rectangular Window Coefficients ---');

disp(b_rect);

%% --- 2. FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm);

%% --- 3. FIR Low Pass using Hanning Window ---

b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');

disp(b_hann);
```

```
%% --- Plot Frequency Responses ---
```

```
figure;
```

```
freqz(b_rect,1);
```

```
title('Low-pass FIR using Rectangular Window');
```

```
figure;
```

```
freqz(b_hamm,1);
```

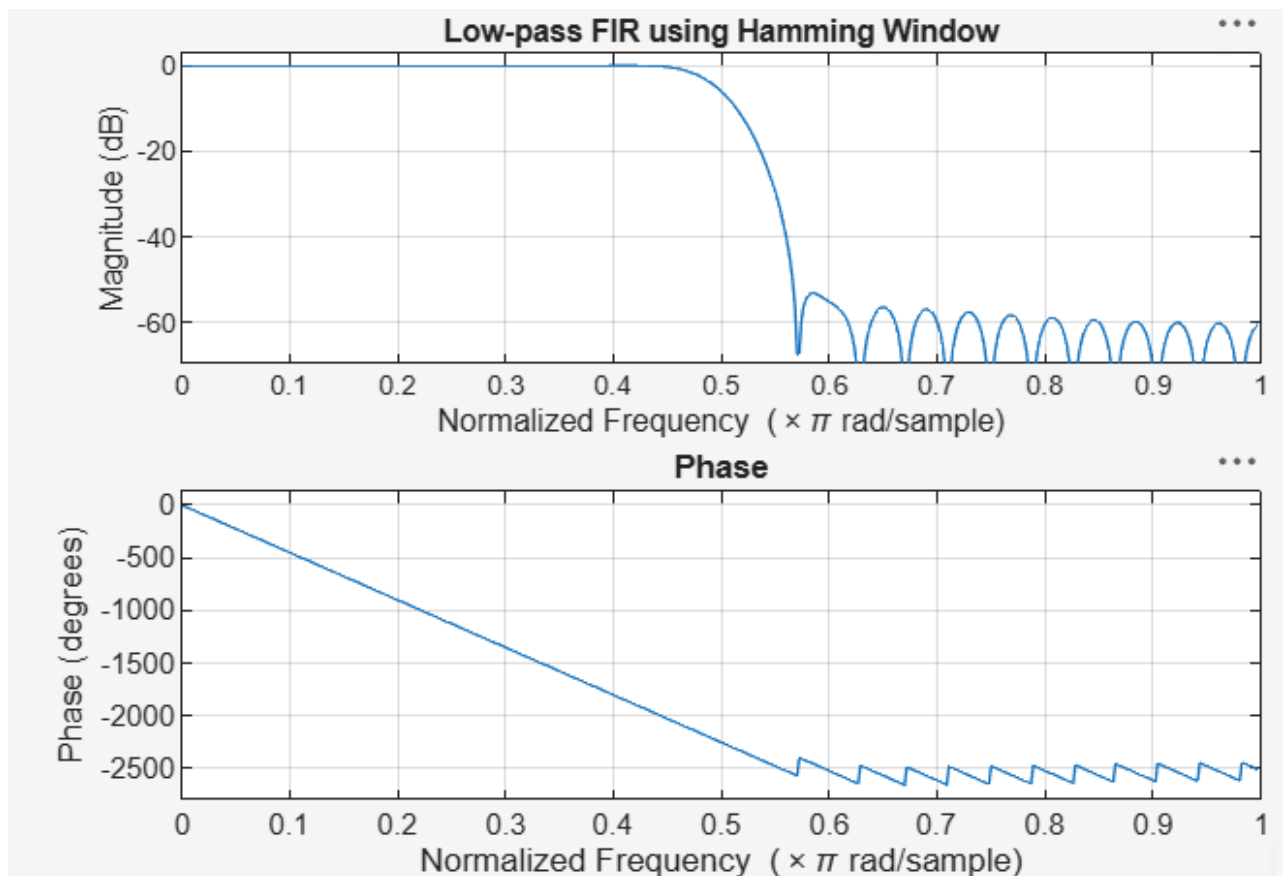
```
title('Low-pass FIR using Hamming Window');
```

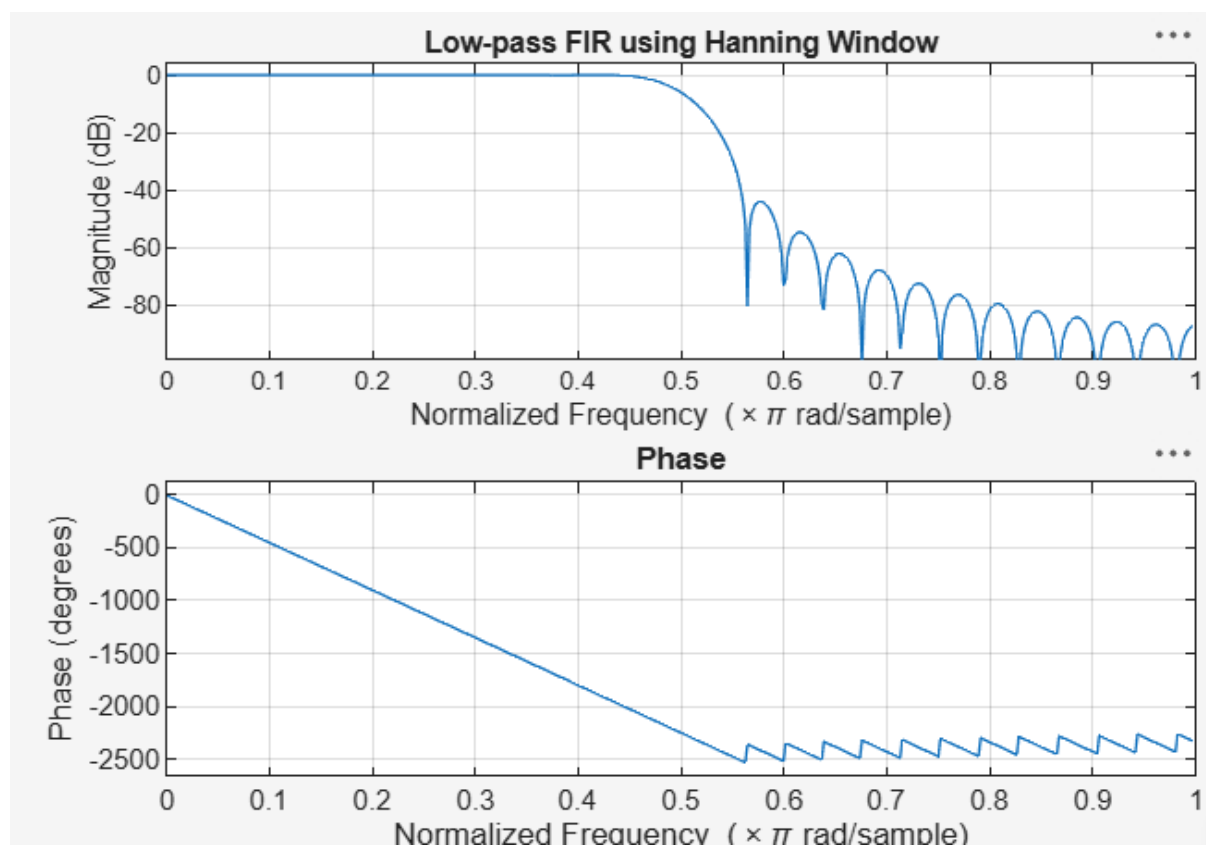
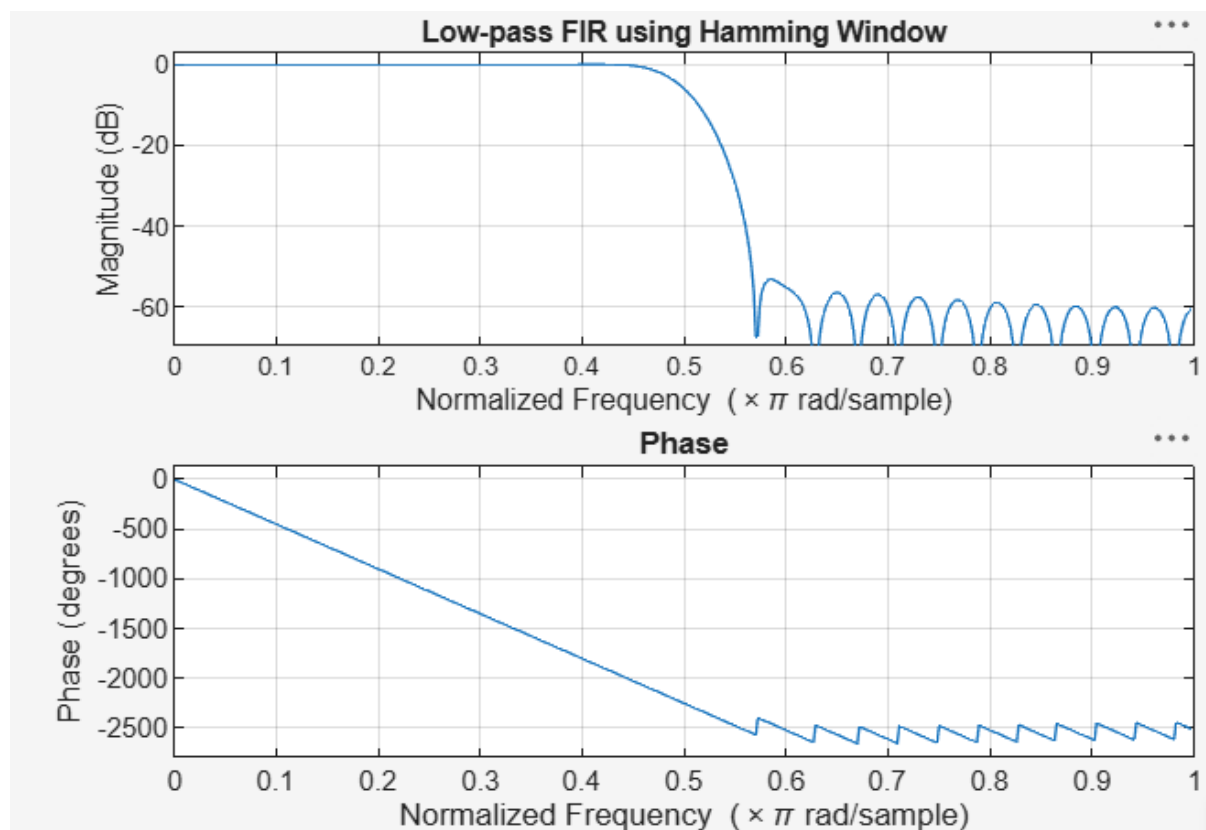
```
figure;
```

```
freqz(b_hann,1);
```

```
title('Low-pass FIR using Hanning Window');
```

OUTPUT





4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB

PROGRAM

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm);

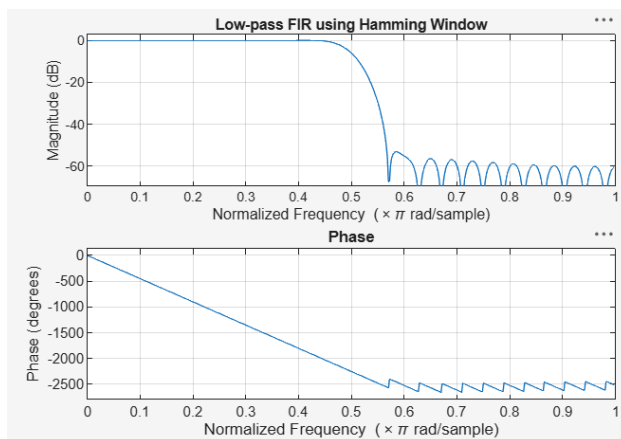
%% --- Plot Frequency Responses ---

figure;

freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window');
```

OUTPUT



4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB

PROGRAM

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann);
%% --- Plot Frequency Responses ---
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

OUTPUT

